

Computer Science & IT

Discrete Mathematics



Combinatorics

Lecture No. 07



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Recap of Previous Lecture



Topic

Generating functions



Topic

Practice questions



Topics to be Covered



Topic

Solution of recurrence relation using
function

generating



Topic

Pigeonhole principle



$$a_k - 3a_{k-1} = 0 \Rightarrow \text{Ch. eqn} = t - 3 = 0, \quad \text{Ch. root} \Rightarrow t = 3$$

$\therefore \text{CF} = C_1(3)^k$ ✓
 $\text{PS} = 0$



#Q. Find the solution of the recurrence relation

✓ $a_k = 3a_{k-1}$, where $k = 1, 2, 3, \dots$ and $a_0 = 2$

$$a_0 = 2, \quad a_1 = 3 \cdot 2 = a_2 = 3 \cdot 3 \cdot 2, \quad a_3 = 3 \cdot 3 \cdot 3 \cdot 2 \dots, \quad a_n = 3^n \cdot 2 = 2 \cdot 3^n$$

$$a_k = 3 \cdot a_{k-1}$$

$$a_0 = 2$$

$$Gf(x) = \sum_{k=0}^{\infty} a_k \cdot (x)^k$$

$$= a_0 x^0 + a_1 x^1 + a_2 x^2 + \dots$$

$$= a_0 x^0 + \sum_{k=1}^{\infty} a_k \cdot (x)^k$$

$$Gf(x) = \underset{\downarrow}{2} \cdot \underset{\downarrow}{x^0} + \sum_{k=1}^{\infty} a_k (x)^k$$

$$Gf(x) = 2 + \sum_{k=1}^{\infty} a_k (x)^k$$

$$= 2 + \sum_{k=1}^{\infty} (3 \cdot a_{k-1}) (x)^k$$

$$= 2 + 3 \cdot \sum_{k=1}^{\infty} a_{k-1} (x)^{k-1} \cdot (x)^1$$

$$= 2 + 3x \sum_{k=1}^{\infty} a_{k-1} x^{k-1}$$

$$Gf(x) = 2 + 3x \cdot \sum_{k=0}^{\infty} a_k x^k$$

$\rightarrow Gf(x)$

$$Gf(x) = 2 + 3x \cdot Gf(x)$$

$$Gf(x) - 3x \cdot Gf(x) = 2$$

$$Gf(x) (1 - 3x) = 2$$

$$Gf(x) = \left(\frac{2}{1 - 3x} \right)$$

$$= 2 * (1 - 3x)^{-1}$$

$$Gf(x) = 2 + \sum_{k=1}^{\infty} a_k (x)^k$$

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$$= 2 + 3x \sum_{k=1}^{\infty} a_{k-1} x^{k-1}$$

$$Gf(x) = 2 + 3x \cdot \sum_{k=0}^{\infty} a_k x^k \rightarrow Gf(x)$$

$$Gf(x) = 2 + 3x \cdot Gf(x)$$

$$Gf(x) - 3x \cdot Gf(x) = 2$$

$$Gf(x) (1 - 3x) = 2$$

$$Gf(x) = \left(\frac{2}{1-3x} \right)$$

$$= 2 * (1-3x)^{-1}$$

$$\therefore a_k = 2 \cdot 3^k$$

$$= 2 * (1-3x)^{-1}$$

Coefficient of x^k

$$= (-3)^k \cdot {}^{n-k}C_k$$

$$= (-3)^k \cdot (-1)^k \cdot {}^{1+k-1}C_k$$

$$= (3)^k \cdot {}^k C_k$$

$$= (3)^k$$



Topic : Pigeonhole principle

→ Let 'A' is the average no. of Pigeons per Pigeonhole

$$A = \frac{\text{No. of Pigeons}}{\text{No. of Pigeonholes}}$$

∴ Some Pigeonhole will contain at least $\lceil A \rceil$ Pigeons
And some Pigeonhole will contain at most $\lfloor A \rfloor$ Pigeons

→ Let $(2n+1)$ Pigeons are distributed in
' n ' Pigeonholes.

$$\text{then, } A = \frac{2n+1}{n} = 2 + \frac{1}{n} \Rightarrow \begin{array}{l} [A] = 3 - (\underbrace{2}_{\downarrow} + 1) \\ [A] = 2 - (\underbrace{2}_{\downarrow} + 0) \end{array}$$

i.e, Some Pigeonhole will contain at least '3' Pigeons

And Some Pigeonhole will contain at most '2' Pigeons

→ If $(kn+1)$ Pigeons are distributed in 'n' Pigeonholes,
then $A = \frac{kn+1}{n} = k + \frac{1}{n} \Rightarrow \lceil A \rceil = k+1$
 $\lfloor A \rfloor = k$

• Some Pigeonhole will contain at least $(k+1)$ Pigeons

And Some Pigeonhole will contain at most $(k+0)$ Pigeons

Q:- Suppose we have 'n' Pigeonholes,

$(kn+1)$ ✓

then what is the minimum number of Pigeons required such that

- (i) some Pigeonhole Contains at least 'k' Pigeons $\xrightarrow{\text{Ans} = \{(k-1) \cdot n + 1\}}$
- (ii) some Pigeonhole Contains at least $(k+1)$ Pigeons. $\xrightarrow{\text{Ans} = (k \cdot n + 1)}$

$$\begin{aligned} \lceil A \rceil = k &= (k-1) + 1 \\ &= ((k-1)n + 1) \end{aligned}$$

MSQ

Students are Pigeons
& '12' months are Pigeonholes } $\therefore A = \frac{\# \text{ Pigeons}}{\# \text{ Pigeonholes}}$



#Q. Consider a group of 61 students, which of the following is/are true?

$\lceil A \rceil = 6$, $\therefore$ There are some months in which
at least 6 students were born

4 $\lfloor A \rfloor = 5$, $\therefore$ There are some months in which at most 5

☒ A) At least 6 students are born in same month.

Students
were born

$$\left\{ \begin{array}{l} A = \frac{61}{12} \\ \lceil A \rceil = 6 \\ \lfloor A \rfloor = 5 \end{array} \right.$$

$D \Rightarrow B$

☒ B) At most 6 students are born in same month.

$\therefore B$ is also true

$A \Rightarrow C$

☒ C) At least 5 students are born in same month.

$\therefore C$ is also true

☒ D) At most 5 students are born in same month.

Pigeonholes $\Rightarrow$ 0° 0 '7' Pigeonholes

#Q. Suppose there are 7 branches in a college and each branch has exactly 50 students, what must be the minimum number of students chosen such that there are at least 10 students from at least one branch?

To ensure that at least $(k+1)$ Pigeons in some Pigeon-hole
We need minimum $(kn+1)$ Pigeon.

→ To ensure $(9+1)$ students from '1' out of '7' branches
We need minimum $(9 \times 7 + 1) = 64$ Students



Upto '63' students it may be the case that we don't have '10' students from same branch

But if no. of students = $(63+1) = 64$, then it is guaranteed that we will have '10' students from at least '1' of those 7 branches

#Q. Suppose we have 8 red color balls, 6 blue color balls, 19 green color balls and 10 yellow color balls, then what must be the minimum number of balls chosen such that we have at least 5 balls of same color?

Red Blue Green Yellow
 ↓ ↓ ↓ ↓
 4 4 4 4

Upto '16' balls we may not have '5' balls of same color

But if we choose $(16+1)=17$ balls then,
 we will have at least '5' balls of same color



2 mins Summary



Topic

Solution of recurrence relation using *generating function*

Topic

Pigeonhole principle

THANK - YOU